

Equality of the two continuous Fourier–Stieltjes norms for discrete groupoids

Automated mathematics run `fa_banach.001`, lane 13

23 July 2026

Status. Complete theorem for the discrete subclass of the source question; current verdict: likely valid. The nondiscrete problem remains open.

1 Source question and result

Let G be a locally compact groupoid with a Haar system. Paterson [1, §5] defines the continuous Fourier–Stieltjes algebra $B(G)$ from coefficients of continuous G -Hilbert bundles. With the convention that Hilbert-space inner products are conjugate-linear in the first variable, a coefficient has the form

$$\phi(g) = (L_g \xi(s(g)), \eta(r(g))),$$

where ξ, η are bounded continuous sections, and

$$\|\phi\|_B = \inf\{\|\xi\|_\infty \|\eta\|_\infty : \phi = (\xi, \eta)\}.$$

Pointwise multiplication by ϕ extends to a completely bounded map

$$T_\phi : C^*(G) \longrightarrow C^*(G), \quad T_\phi(f) = \phi f,$$

and one sets $\|\phi\|_{\text{cb}} = \|T_\phi\|_{\text{cb}}$. Paterson records the always-valid inequality

$$\|\phi\|_{\text{cb}} \leq \|\phi\|_B \tag{1}$$

and asks whether equality holds in general. The source gives equality for the trivial pair groupoid R_n , as well as a separate proof for locally compact abelian groups.

The exact source sentence (Section 5, page 10 of the paper) is:

“It is an open question if the two norms actually coincide in general.”

The surrounding source passage is reproduced below.

sections of continuous Hilbert bundles, every $\phi \in B(G)$ is a linear combination of continuous positive definite elements of $B(G)$ and conversely. From the discussion in §4, we see that $\mathcal{B}(G) \cap C(G) \neq B(G)$ in general. Intuitively, $\mathcal{B}(G)$ captures the representation theory of G but we need to consider Borel functions. On the other hand, $B(G)$ may not capture all of the representation theory of G (though it does capture at least the regular representation) but has the advantage that the functions involved, are, as in the group case, continuous. (Our $B(G)$ is the same as Oty's $B_1(G)$ ([61]).)

We defined the norm on $B(G)$ using Hilbert bundles in a way similar to that in which Renault defined the norm on $B_\mu(G)$, and it is natural to enquire if the method of Ramsay and Walter for norming $\mathcal{B}(G)$ can also be applied in the present context. This is in fact the case. The difference is that we regard $\phi \in B(G)$ as a multiplier on $C_c(G)$, this action extending to a completely bounded map T_ϕ on $C^*(G)$ instead of on $M^*(G)$. We can then define a norm $\|\cdot\|_{cb}$ on $B(G)$ by taking $\|\phi\|_{cb} = \|T_\phi\|_{cb}$. Following along the same lines as the corresponding argument in [69], one can show that $B(G)$ is a Banach algebra under $\|\cdot\|_{cb}$. Since it is not difficult to show that $\|\cdot\|_{cb} \leq \|\cdot\|$, it follows by Banach's isomorphism theorem that the two norms are equivalent on $B(G)$. It is an open question if the two norms actually coincide in general. It is shown, using a result of Paulsen on Schur multipliers ([66]) that they do coincide when $G = R_n$, the trivial groupoid on n elements (§2).

Let us sketch here how one can show that the two norms are the same when G is a locally compact abelian group. As in §1, we can regard ϕ as a measure $\mu \in M(\hat{G})$. Recalling that $C^*(G) = C_0(\hat{G})$, the map T_ϕ goes over to the map

The theorem below supplies the reverse inequality throughout the discrete case. As usual in the source setting, “discrete” plus second countability means countable; equivalently, one can take countability as an explicit hypothesis.

2 Proof intuition

The cb multiplier itself contains the desired coefficient factorization once it is observed through the trivial one-dimensional groupoid representation. Indeed, that representation sends an arrow $g : v \rightarrow u$ to the matrix unit $E_{u,v}$. A Haagerup–Paulsen factorization of the resulting operator-valued map can then be cut by the mutually orthogonal unit projections of $C^*(G)$. Those cuts are the Hilbert fibers, the images of the arrow point masses are the fiberwise unitaries, and the columns of the two factorizing operators are bounded coefficient sections. This reconstructs ϕ at cost at most $\|T_\phi\|_{cb}$, precisely reversing (1). Discreteness is what makes the point masses and the pointwise fiber columns available; the nondiscrete source question remains open.

Theorem 1. *Let G be a countable discrete groupoid, equipped with its counting Haar system. Then*

$$\|\phi\|_B = \|T_\phi\|_{cb} \quad \text{for every } \phi \in B(G).$$

3 The trivial line-bundle representation

Write G^0 for the unit space and put

$$A = C^*(G), \quad H_0 = \ell^2(G^0).$$

For $u \in G^0$, let e_u be the corresponding unit vector. For $g \in G$, write $\delta_g \in C_c(G)$ for the point mass and $p_u = \delta_u$.

Lemma 2. *There is a $*$ -representation $\sigma : A \rightarrow B(H_0)$ satisfying*

$$\sigma(\delta_g)e_{s(g)} = e_{r(g)}, \quad \sigma(\delta_g)e_v = 0 \quad (v \neq s(g)).$$

Equivalently, $\sigma(\delta_g) = E_{r(g),s(g)}$.

Proof. Give every unit the one-dimensional Hilbert fiber $\mathbb{C}$, and let every arrow act by the identity map between its source and range fibers. This is the trivial G -Hilbert bundle. Its integrated representation on $\ell^2(G^0)$ has the displayed formula.

One can also check the formula directly. If g, h are composable, then

$$E_{r(g),s(g)}E_{r(h),s(h)} = E_{r(gh),s(gh)};$$

if they are not composable, the product is zero, and adjoints correspond to inverses. Thus the assignment is a $*$ -representation of $C_c(G)$. For $f \in C_c(G)$, its matrix row and column sums are bounded by the two terms in the groupoid I -norm, so the assignment is I -norm bounded. It therefore extends to the full groupoid C^* -algebra. $\square$

The fact that distinct arrows with the same source and range are sent to the same matrix unit causes no loss: the domain elements δ_g remain distinct, and we will evaluate the resulting operator-valued map separately on every one of them.

4 Reconstructing a coefficient from a cb factorization

We use the standard Haagerup–Paulsen factorization theorem [2, Ch. 8]: if $S : A \rightarrow B(H_0)$ is completely bounded, then for each $\varepsilon > 0$ there are a Hilbert space K , a $*$ -representation $\pi : A \rightarrow B(K)$, and operators $V, W : H_0 \rightarrow K$ such that

$$S(a) = V^*\pi(a)W, \quad \|V\| \|W\| \leq \|S\|_{\text{cb}} + \varepsilon. \quad (2)$$

Lemma 3. *For a countable discrete groupoid G and every $\phi \in B(G)$,*

$$\|\phi\|_B \leq \|T_\phi\|_{\text{cb}}.$$

Proof. Fix $\varepsilon > 0$ and compose the multiplier with the representation from Lemma 2:

$$S = \sigma \circ T_\phi : A \longrightarrow B(H_0).$$

Then $\|S\|_{\text{cb}} \leq \|T_\phi\|_{\text{cb}}$. Choose a factorization (2).

The mutually orthogonal projections $\pi(p_u)$ define Hilbert fibers

$$K_u = \pi(p_u)K \quad (u \in G^0).$$

For every arrow g , the restriction

$$U_g = \pi(\delta_g)|_{K_{s(g)}} : K_{s(g)} \longrightarrow K_{r(g)}$$

is unitary, because

$$\delta_g^* \delta_g = p_{s(g)}, \quad \delta_g \delta_g^* = p_{r(g)}.$$

The convolution relations give $U_g U_h = U_{gh}$ for composable arrows. Thus (K_u, U_g) is a G -Hilbert bundle. Since G^0 is discrete, there is no continuity obstruction.

Define bounded sections

$$\xi(u) = \pi(p_u)W e_u, \quad \eta(u) = \pi(p_u)V e_u.$$

They satisfy

$$\|\xi\|_\infty \leq \|W\|, \quad \|\eta\|_\infty \leq \|V\|. \quad (3)$$

For $g \in G$, using Paterson's convention that the inner product is conjugate-linear in the first variable, we obtain

$$\begin{aligned} (U_g \xi(s(g)), \eta(r(g))) &= (\pi(\delta_g)W e_{s(g)}, V e_{r(g)}) \\ &= (V^* \pi(\delta_g)W e_{s(g)}, e_{r(g)}) \\ &= (S(\delta_g) e_{s(g)}, e_{r(g)}) \\ &= (\phi(g) e_{r(g)}, e_{r(g)}) = \overline{\phi(g)}. \end{aligned}$$

Hence $\overline{\phi}$ is a coefficient with the bounds (3). Passing to the conjugate Hilbert bundle turns this coefficient into ϕ : if $(\overline{x}, \overline{y})_{\overline{K}} = (y, x)_K$, then

$$(\overline{U_g \xi(s(g))}, \overline{\eta(r(g))})_{\overline{K}} = \overline{(U_g \xi(s(g)), \eta(r(g)))_K} = \phi(g).$$

Conjugation preserves all fiber norms. Consequently,

$$\|\phi\|_B \leq \|\xi\|_\infty \|\eta\|_\infty \leq \|V\| \|W\| \leq \|S\|_{\text{cb}} + \varepsilon \leq \|T_\phi\|_{\text{cb}} + \varepsilon.$$

Letting $\varepsilon \downarrow 0$ proves the lemma. □

Proof of Theorem 1. Combine Lemma 3 with the established inequality (1). □

Corollary 4. *The two norms coincide for every finite groupoid.*

Remark 5 (What this adds to the recorded finite case). A transitive finite groupoid is isomorphic, after choices of connecting arrows, to

$$I \times H \times I,$$

where H is an isotropy group; its full C^* -algebra is $M_{|I|}(C^*(H))$. Paterson's R_n is the case $H = \{e\}$. Corollary 4 allows arbitrary finite H , including nonabelian isotropy, and arbitrary finite disjoint unions of such components.

5 Scope audit and novelty check

Why discreteness enters

The proof uses both the atomic vectors $e_u \in \ell^2(G^0)$ and the isolated arrow functions $\delta_g \in C_c(G)$. They let a single completely bounded map recover the value $\phi(g)$ arrow by arrow. For a nondiscrete unit space, counting measure need not be Radon and the constructed fiber vectors need not form continuous sections. Thus the proof does not establish the general locally compact statement.

Adversarial checks

Potential failure	Check
Parallel arrows collapse under σ	Only the output matrix unit repeats; S is evaluated on each distinct domain basis element δ_g , so its scalar is still $\phi(g)$.
The factorizing π is not a bundle representation	The exact relations for p_u, δ_g cut K into fibers and make every $\pi(\delta_g)$ a unitary between the correct fibers.
Sections are not bounded or continuous	Projection gives the uniform estimates (3); every section over a discrete base is continuous.
Inner-product convention conjugates the answer	The first coefficient is $\overline{\phi}$; the conjugate Hilbert bundle gives ϕ with identical norms.
Factorization estimates the wrong norm	σ is a $*$ -representation, so $\ \sigma T_\phi\ _{\text{cb}} \leq \ T_\phi\ _{\text{cb}}$.

Bounded literature search

On 23 July 2026, searches were run for the exact source question and for combinations of *discrete groupoid*, *r-discrete groupoid*, *Fourier–Stieltjes*, *coefficient norm*, and *completely bounded multiplier*. Oty’s earlier work [3] studies the Borel/r-discrete Fourier–Stieltjes algebra with a norm defined through multipliers on the Borel completion $M^*(G)$; this is not a statement of the equality proved here for Paterson’s continuous Hilbert-bundle coefficient norm on $B(G)$. Later operator-space duality [4] is in the measured setting, while recent restriction-map work [5] addresses a different problem. No exact prior statement was located. This is a bounded negative search, so novelty confidence is moderate rather than definitive.

Human-review recommendation

Send to a specialist reviewer. The two points deserving the closest check are the application of Haagerup–Paulsen factorization to σT_ϕ and the conjugate-bundle calculation. If accepted, the theorem is suitable for a concise proposition or short note.

References

- [1] A. L. T. Paterson, *The Fourier–Stieltjes and Fourier algebras for locally compact groupoids*, arXiv:math/0310138 (2003).
- [2] V. I. Paulsen, *Completely Bounded Maps and Operator Algebras*, Cambridge Studies in Advanced Mathematics 78, Cambridge University Press, 2002.
- [3] K. J. Oty, *Fourier–Stieltjes algebras of r-discrete groupoids*, J. Operator Theory **41** (1999), 175–197.
- [4] A. L. T. Paterson, *An operator space duality theorem for the Fourier–Stieltjes algebra of a locally compact groupoid*, arXiv:1102.0304 (2011).
- [5] J. DeGaetani and M. Ghandehari, *On the restriction maps of the Fourier and Fourier–Stieltjes algebras over locally compact groupoids*, Int. Math. Res. Not. IMRN (2025), rnaf057.